

UNIT – 2 TWO DIMENSIONAL RANDOM VARIABLES TUTORIAL SHEET

1. A fair coin is tossed four times. Let X denote the number of heads obtained in the first two tosses, and let Y denote the number of tails obtained in the last two tosses. Find the joint PMF of X and Y.

2. The joint probability density function of the random variable (X,Y) is given as

$$f(x,y) = \begin{cases} e^{-(x+y)}; & x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}. \text{ Check whether the random variables X and Y are independent or not?}$$

3. Show that the correlation co-efficient is always lies between -1 and +1.

4. Two random variables X and Y have the joint probability density function given by $f_{XY}(x,y) = \begin{cases} \frac{1}{2}e^{-2y}; & 0 < x < 4, y > 0 \\ 0 & \text{otherwise} \end{cases}$. Determine the marginal density functions of X and Y.

5. Two independent resistors X and Y are connected in parallel in a network and are uniformly distributed in the interval 90 and 100 ohms. Find the probability density function of the net resistance of the network.

6. If the joint probability density function of (X,Y) is given by $f_{XY}(x,y) = \begin{cases} 2 & ; 0 < x < y < 1 \\ 0 & ; \text{otherwise} \end{cases}$.

Find the regression lines X on Y and Y on X.

7. Obtain the correlation coefficient from the following data:

x:	22	26	29	30	31	31	34	35
y:	20	20	21	29	27	24	27	31

8. Obtain the equations of the lines of regression from the following data:

X:	1	2	3	4	5	6	7
Y:	9	8	10	12	11	13	14

9. The two regression lines are $4X-5Y+33=0$ and $20X-9Y-107=0$. Find the regression coefficients.

10. If x and Y are independent exponential random variables with parameters 2 and 3 respectively, find the density function of their sum X+Y.

11. Determine whether the random variables X and Y are independent, given their joint probability density function as

$$f(x,y) = \begin{cases} x^2 + \frac{xy}{3}, & 0 \leq x \leq 1, 0 \leq y \leq 2 \\ 0, & \text{otherwise} \end{cases}.$$

12. Let (X,Y) be two dimensional random variable described by the joint p.d.f

$$f(x,y) = \begin{cases} 2-x-y & ; 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & ; \text{otherwise} \end{cases}$$

Find the correlation coefficient between X and Y.

13. Two random variables X and Y are defined with $Y=4X+9$. Find the correlation coefficient between X and Y.

14. Two random variables X and Y have the joint probability density function

$$f(x,y) = \begin{cases} \frac{6}{7}x; & 1 \leq x+y \leq 2, x > 0, y > 0 \\ 0 & ; \text{otherwise} \end{cases}$$

Obtain the two regression lines X on Y and Y on X.

15. Find the PDF of Y in terms of the PDF of X if $Y=2X+7$.

16. Find the PDF of the random variable $Y=X^2$, where X is the standard normal random variable.

17. Find the PDF of the random variables X+Y, X-Y, XY, X/Y and X^2 where the joint PDF of X and Y, $f_{XY}(x,y)$ is given.

18. If 30 fair dice are rolled, what is the probability that the sum obtained is between 92 and 125?

19. $X_1, X_2, \dots, X_{35}$ are independent random variables each of which is uniformly distributed between 0 and 1. Let $S = X_1 + X_2 + \dots + X_{35}$. What is the probability that $S > 22$?

20. Suppose we select one point at random from within the circle with radius R. Let the centre of the circle denote the origin and define X and Y to be the coordinates of the point chosen. Find the marginal pdf's of X and Y. Find the probability that the distance from the origin of the point selected is not greater than a.

21. The time X between consecutive snowstorms in winter is a exponential random variable with parameter λ . Assume it has not snowed up until now. What is the PDF of the time U until the second snowstorm?

22. Find the mean of the two variables involved given that the coefficient of correlation between the variables is -1/2, one line of regression passes through the points (4,0) and (-14,3) and the other line of regression passes through the point (1,-1).

23. Coefficient of correlation between X and Y is 0.48. Their covariance is 36. The variance of X is 16, find the standard deviation of Y.

24. Let X and Y are independent random variables follows exponential distribution with mean unity and uniform distribution in (-1,1) respectively. Find the density function of X+Y.

25. The random variable X is uniformly distributed in the interval (0,2), with $F_X(0.3)=0.4$. Find k such that $F_Y(k)=0.6$ where $Y=1-X$.